

Does Random Mean Secure?

Randomness, Entropy and Perfect Secrecy

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September 13, 2026

Two meanings of “random”

1. A sequence *looks* random: balanced, right run lengths, nothing repeats. (statistics)
2. A sequence *is* unpredictable: knowing everything else, you still cannot guess the next bit. (cryptography)

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The question

If the randomness behind a key gets worse, how much more does the ciphertext reveal about the message?

Shannon (1949): a cipher is perfectly secret if for every m and c

$$\mathbb{P}(M = m \mid C = c) = \mathbb{P}(M = m).$$

Equivalently, $I(M; C) = H(M) - H(M \mid C) = 0$.

For $C = M \oplus K$ with K uniform on $\{0, 1\}^n$ and independent of M :

$$\mathbb{P}(C = c \mid M = m) = \mathbb{P}(K = m \oplus c) = 2^{-n},$$

which does not depend on m , so Bayes' rule gives perfect secrecy.

The main experiment

Fix the cipher $C = M \oplus K$, $M \in \{0,1\}^2$ lopsided. Only change the key source.

Key	Expected $I(M; C)$
A. uniform	$\rightarrow 0$
B. one value w.p. 0.7	> 0
C. bits 1 w.p. $q \in [0.5, 0.95]$	grows with q

Main figure: leakage against key bias, simulation vs. exact.

Good randomness is necessary for secrecy.
But looking random is not enough to make something secure.